SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	**ITA0512**	**Course Title:**	**Computer Vision**
CO Assessed:	**CO4 – Articulation (combining methods for evaluation) P4**	**Assessment:**	**Assessment Tool – Comparative Analysis**
Weightage:	**10% of CIA**	**Total Marks:**	**100**
CO4 Statement:	*Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)*		
Student Name:	*S. Susannal Devapriyam*	**Reg. No.:**	*192525398*
Section / Batch:	B. Tech AI & ML	**Date:**	26 - 08 -2026

Code of Conduct

I, S. Susannal Devapriyam (192525398) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *S. Susannal Devapriyam*

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**	**Marks**
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Questions :

10. Error Rate Comparison

Results:
Method Errors Total
A 20 100
B 10 100
(a) Compute error rate.
(b) Compare and evaluate.

11. Motion Estimation Variance

Errors:
Frame A B
1 2 6
2 3 7
3 2 5
(a) Compute mean error.
(b) Compare variability and evaluate stability.

12. IOU Comparison

IOU values:
Object A B
1 0.6 0.8
2 0.7 0.85
3 0.65 0.9
(a) Compute average IOU.
(b) Compare and evaluate detection quality.

Answers :

10. Error Rate Comparison

Given Data

Method	Errors	Total Samples
A	20	100
B	10	100

(a) Compute Error Rate:

The error rate is calculated using:

Error Rate = (Number of Errors / Total Samples) × 100

Method A

Error Rate = (20 / 100) × 100 = **20%**

Method B

Error Rate = (10 / 100) × 100 = **10%**

(b) Compare and Evaluate:

Method	Error Rate	Performance
A	20%	Higher error
B	10%	Lower error

- Method A produces errors in 20 out of every 100 samples.
- Method B produces errors in only 10 out of every 100 samples.
- Method B has **half the error rate** of Method A.
- Therefore, Method B demonstrates better accuracy and reliability.
- Based on the given results, **Method B should be preferred**.

Conclusion

Method B is superior because its 10% error rate is significantly lower than Method A's 20% error rate.

11. Motion Estimation Variance

Given Data

Frame	Method A Error	Method B Error
1	2	6
2	3	7
3	2	5

(a) Compute Mean Error:

Method A

Mean Error = (2 + 3 + 2) / 3

= 7 / 3

= **2.33**

Method B

Mean Error = (6 + 7 + 5) / 3

= 18 / 3

= **6**

Mean Error Comparison

Method	Mean Error
A	2.33
B	6.00

Method A has a considerably lower mean error than Method B.

(b) Compare Variability and Evaluate Stability:

Variance is used to measure how much the error values vary from their mean.

Method A Variance

Mean = 2.33

Error	Deviation	Squared Deviation
2	-0.33	0.11
3	0.67	0.44
2	-0.33	0.11

Variance = (0.11 + 0.44 + 0.11) / 3

= **0.22**

Method B Variance

Mean = 6

Error	Deviation	Squared Deviation
6	0	0
7	1	1
5	-1	1

Variance = (0 + 1 + 1) / 3

= **0.67**

Comparison

Method	**Mean Error**	**Variance**	**Stability**
A	2.33	0.22	High
B	6.00	0.67	Lower

Evaluation

- Method A has the lower mean error of **2.33**.
- Method B has a higher mean error of **6.00**.
- Method A also has lower variance, indicating more consistent performance.
- Method B shows greater variation between different frames.
- Therefore, Method A is more accurate and stable for motion estimation**.**

Conclusion

Method A is preferred because it provides both lower average error and lower variability, making its motion estimation more reliable.

12. IOU Comparison

Given Data

Object	Method A	Method B
1	0.60	0.80
2	0.70	0.85
3	0.65	0.90

(a) Compute Average IOU:

Method A

Average IOU = (0.60 + 0.70 + 0.65) / 3

= 1.95 / 3

= **0.65**

Method B

Average IOU = (0.80 + 0.85 + 0.90) / 3

= 2.55 / 3

= **0.85**

(b) Compare and Evaluate Detection Quality:

(c)

Method	Average IOU	Detection Quality
A	0.65	Moderate
B	0.85	High

Evaluation

- IOU measures the overlap between predicted and actual object regions.
- A higher IOU indicates more accurate object localization.
- Method B achieves higher IOU values for all three objects.
- Method B has an average IOU of **0.85**, compared to **0.65** for Method A.
- Therefore, Method B provides more accurate and reliable object detection.

Conclusion

Method B demonstrates better detection quality because its predicted bounding regions have greater overlap with the ground truth objects.